Practical Training of Deep Networks

Initialisation, normalisation, clipping, schedules, and diagnosis

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Training is a coupled dynamical system

Initialisation sets the starting signal scale

Variance propagation through deep networks

Normalisation changes optimisation geometry

Batch normalisation in training and evaluation

Residual pathways preserve useful signals

Gradient clipping limits unstable updates

Learning-rate schedules control training phases

Training curves are diagnostic evidence

Inspecting activations and gradients

A systematic failure-diagnosis workflow